

Problem 1: Finding Complexity using Counter Method

Started on	Monday, 3 August 2026, 10:22 AM
State	Finished
Completed on	Monday, 3 August 2026, 10:45 AM
Time taken	22 mins 46 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00 | [Flag question](#)

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

```

1  #include<stdio.h>
2  int main(){
3      int n;
4      int i=1,s=1;
5
6      int count=0;
7      count++;
8      scanf("%d",&n);
9      count++;
10     while(s<=n)
11     {
12         count++;
13         |
14         i++;
15         count++;
16         s+=i;
17         count++;
18

```

```

15         count++;
16         s+=i;
17         count++;
18
19     }
20     count++;
21     printf("%d",count);
22     count++;
23 }

```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



Problem 2: Finding Complexity using Counter method

Started on	Monday, 3 August 2026, 10:45 AM
State	Finished
Completed on	Sunday, 9 August 2026, 8:33 PM
Time taken	6 days 9 hours
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00 | [Flag question](#)

Question 1 | Correct | Mark 1.00 out of 1.00 | [Flag question](#)

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 |
3 int main()
4 {
5     int n;
6     int count;
7
8     scanf("%d", &n);
9
10    if (n == 1)
11        count = 2;
12    else
13        count = 5 * n + 2;
14
15    printf("%d", count);
16    return 0;
17 }
18 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



Problem 3: Finding Complexity using Counter Method

Started on Monday, 3 August 2026, 11:31 AM

State Finished

Completed on Monday, 3 August 2026, 11:35 AM

Time taken 3 mins 18 secs

Marks 1.00/1.00

Grade 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00 [Flag question](#)

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {  
  {  
    for (i = 1; i <= num; ++i)  
    {  
      if (num % i == 0)  
      {  
        printf("%d ", i);  
      }  
    }  
  }  
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>  
2  int main(){  
3      int num;  
4      scanf("%d",&num);  
5      int count=0;  
6      for(int i=1;i<=num;i++){  
7          count++;  
8          if(num%i==0){  
9              count++;  
10             }  
11         }  
12         count++;  
13     }  
14     count++;  
15     printf("%d",count);  
16 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



Problem 4: Finding Complexity using Counter Method

Started on	Monday, 3 August 2026, 11:35 AM
State	Finished
Completed on	Monday, 3 August 2026, 11:38 AM
Time taken	3 mins 23 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00 | [Flag question](#)

Convert the following algorithm into a program and find its time complexity using counter method.

Question 1 | Correct | Mark 1.00 out of 1.00 | [Flag question](#)

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int c=0;
6      int count=0;
7      count++;
8      for(int i=n/2;i<n;i++){
9          count++;
10         for(int j=1;j<n;j=2*j){
11             count++;
12             for(int k=1;k<n;k=k*2){
13                 count++;
```

```

11     count++;
12     for(int k=1;k<n;k=k+2){
13         count++;
14         c++;
15         count++;
16     }
17     count++;
18 }
19 count++;
20 }
21 count++;
22 printf("%d",count);
23 }

```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

CS23551-PAA-2023-CSE / Problem 5: Finding Complexity using counter method



Problem 5: Finding Complexity using counter method

Started on	Monday, 3 August 2026, 11:38 AM
State	Finished
Completed on	Sunday, 9 August 2026, 8:41 PM
Time taken	6 days 9 hours
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00 | [Flag question](#)

Convert the following algorithm into a program and find its time complexity using counter method.

```

void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;
    }
    print(rev);
}

```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int rev=0;
6
7      int rem;
8      int count=0;
9      count++;
10     count++;
11     while(n!=0){
12         count++;
13         rem = n % 10;
14         count++;
15         rev = rev * 10 + rem;
16         count++;
17         n/= 10;
18         count++;
19     }
20     count++;
21     printf("%d",count);
22 }
```

```
20     count++;
21     printf("%d",count);
22 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.